

Case Study - Dropline Four Connect

A familiar board game with durable branching studies, atomic subtree reuse and bounded tactical proofs

Harbor task: turing/dropline-four-connect | version 1.0.0 | prepared 14 September 2026

Recorded platform outcomes on the frozen r6 package. No score adjustment or fresh paid run was performed during delivery preparation. Interpretation limits are stated below.

1. What the task is

Build a complete two-person Connect Four web application with private saved games and analysis tools. The visible game must remain consistent with SQLite through undo, redo, sign-out, concurrent tabs and server restart. A static board or attractive front end alone is insufficient.

Feature area	Required behavior
Identity and durable games	Two seeded accounts; protected bearer sessions and account-wide logout; exact boards, scores, histories, revisions and private archives.
Connect Four and replay	Gravity, full columns, wins in every direction and both colors, exact draw, terminal lock, reversible scores, redo branching, latest-ten archive and read-only replay.
Private branching studies	Fork any replay position; frozen source snapshot; nested alternatives, child reuse, rename, selection, comparison and competitive-score isolation.
Transactional subtree transplant	Preview every derived position, recursive identity-preserving merges, same-study frozen traversal, illegal-descendant rollback, dual-revision conflicts and durable retry receipts.
Bounded tactical proofs	Depths 1-4, terminal-before-horizon evaluation, adversarial win/loss/draw/unknown outcomes, shortest wins, longest resistance, complete ordered proof and browsable board positions.
Persistence and presentation	Two process restarts, SQLite state and receipt durability, input rejection/ownership; keyboard/focus, pending feedback, reduced motion and 375px layouts; six Visual axes.

2. What is verified

58 independently reported criteria across five batched dimensions: Render 1, Constraints 2, Functional 42, Polish 7 and Visual 6. Functional criterion weights total 65.5. Real UI journeys and captured legitimate requests test behavior; exact root/node fixtures and read-only full-proof traversal avoid random gameplay. Shared setups retain separate verdicts.

Codex judge: gpt-5.6-luna, temperature 0, configured reasoning effort max. Render and Constraints are gates; otherwise Overall = 0.6 x Functional + 0.2 x Polish + 0.2 x Visual. Public networking is enabled for agent and separate verifier. Agent timeout is 7200 seconds; verifier timeout is 13200 seconds.

3. Where the runs landed

Run	Overall	Functional	Polish	Visual	Recorded status
Oracle	0.9917	1.0000	1.0000	0.9583	42/42; both gates pass
GPT-5.4 mini (high)	0.1766	0.0305	0.0000	0.7917	1/42; both gates pass
Gemini 3.7 Flash	0.7804	0.8550	0.5455	0.7917	35/42; both gates pass

Run	Overall	Functional	Polish	Visual	Recorded status
Claude Haiku 4.5	0.0000	0.0000	0.0000	0.5833	0/42; Constraints = 0
No-op	0.0000	0.0000	0.0000	0.0000	graded=0; no_op=1

Render is 1.0 for all four built applications. Constraints is 1.0 for Oracle, GPT-mini and Gemini, and 0.0 for Haiku. No-op zeros are control outputs, not 58 independently observed failures.

Oracle

Oracle scored 0.9917 overall: all 42 Functional criteria passed, with Render, Constraints and Polish also at 1.0. Visual was 0.9583. Only visual_spacing_and_layout received 4/5 rather than 5/5: the judge noted imbalance in the wide completed-matches card and tall analysis workspace. This is not an overall 1.0 result. The prior analysis pending-input and restart observations passed in this run.

GPT-5.4 mini

GPT-mini scored 0.1766 with both gates passed. Only the weight-2 seeded_state_import_and_isolation criterion passed Functional (2/65.5 = 0.0305). The exported UI and manifest omit New game, a required workflow entry point; many later failures are blocked setup, not separately demonstrated algorithm defects. Polish also reports absent replay Previous/Next/Close controls and a TypeError leaving analysis boards and selectors empty.

Gemini 3.7 Flash

Gemini scored 0.7804 and passed 35/42 Functional criteria. Recorded failures concern keyboard Undo/Redo focus, malformed operation identifiers in analysis and transplant commits, tactical defence/loss calculations, full-column recommendations, stale report invalidation and extra report fields. Polish deductions concern clipped mobile analysis, repeated pending commit activation and lost focus/feedback. Its overall score remains above 0.7; only GPT-mini meets the stated below-0.7 target in this comparison.

Claude Haiku 4.5

Haiku scored 0.0000 because local_refresh_and_health failed: the judge observed HTTP 200 health/game responses but a blank authenticated page and a null-status TypeError after reload. This is a Constraints-gated zero, not absence of all working code. Its Functional judge recorded inverted row coordinates, stale winner display, draw-archive Undo errors and missing analysis/transplant/tactical surfaces. Some multi-tab checks used shared storage rather than the requested isolated contexts and were unverified.

4. What the separation demonstrates

The golden solution demonstrates complete Functional coverage under the recorded judge, while GPT-mini fails basic workflow entry points despite a relatively strong Visual score. Gemini implements substantially more behavior but loses specific state-validation, search and interaction checks. This is evidence of distinct implementation outcomes, not proof of stable model rankings across repeated runs.

5. Interpretation limits

- GPT analysis creation uses a native browser prompt for the name. The Functional explanation that the creation name field was hidden does not match that code path; the hidden field is for renaming. A dismissed/unhandled native prompt can produce the reported name-required message. Without exported detailed judge actions, that creation failure is not a cleanly established missing-input defect. Other dimensions report a separate analysis-render TypeError; these are distinct observations.

- GPT cross_tab_sign_out_revocation observed distinct tokens, proactive sign-out and old-token 401 rejection, but post-revocation re-login verification was unfinished after a tool timeout. Do not describe the failed criterion as proof that logout revocation is broken.
- Missing New game and inaccessible studies cause many dependent GPT failures. Their number cannot be interpreted as independent evidence of 41 different algorithm defects. The recorded score is preserved, not corrected or presented as a complete isolated test of its backend.
- Haiku multi-tab evidence was limited by a shared-context reload failure. Its main reload defect is also reported separately by Constraints; the unexecuted distinct-session probes are not additional demonstrated session bugs.
- The exports contain all criterion verdicts and no trial-level exceptions, but that does not mean every blocked or unverified subcheck executed. This delivery is not blanket fairness certification. No local browser regrade of these model submissions was performed during final packaging.
- One current exported attempt per model is included. This is not a repeatability study. Oracle is above 0.95 with Functional 1.0, not overall 1.0; Gemini is above 0.7 and Haiku is a gate-zero. Administrative acceptance is not asserted.

6. Delivery and provenance

The task ZIP, four complete named job archives (Oracle includes no-op), and companion evaluation report are supplied. Current source matches the task ZIP, all exported verifier hashes match it, and every golden app file matches the Oracle app export. Task files and scores were not changed for this delivery.

Task ZIP SHA-256: 577451c9d67ef834db1c8366af284b323233ad4f7abc20e396c23c562dd1d63a